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Cartel detection in public procurement – Evaluation of five econometric methods

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Abstract

Purpose – In this paper, the author tests five econometric methods for cartel detection suggested in the aforementioned papers on a data set covering the so-called Swedish asphalt cartel, where collusive bidding was detected by the Swedish Competition Authority (SCA) in 2001 and later verified by court in 2003. This paper also elaborates on the econometric specifications suggested by Lundberg (2017) and Bergman *et al*. (2020). The analysis concentrates on collusive bidding on the same contract. Finally, this paper relates the author's findings to the analysis by Aaltio *et al.* (2025), who applied the distributional regression approach suggested by Clark *et al.* (2025) and the machine learning approach suggested by Huber and Imhof (2019) to the Swedish asphalt cartel.

Design/methodology/approach – The author's data set contains information on procurements of paving and asphalt works (surfacing works) during the period 1995–2009 and the identities of convicted firms, making it possible to analyze bidding behavior before and after the 2003 court order. The empirical strategy is to first test if the econometric methods suggested by Porter and Zona (1993, 1999) and Bajari and Ye (2003) are able to verify the asphalt cartel, as did the methods suggested by Lundberg (2017) and Bergman *et al.* (2020). This paper also elaborates on the econometric specifications suggested by Lundberg and Bergman *et al.*

Findings – In short, when applied to the Swedish asphalt cartel and in contrast to the findings by Porter and Zona (1993) for highway construction contracts on Long Island in the early 1980s, this paper does not find any systematic differences in bidding behavior between competitive and collusive bidders or evidence supporting the hypothesis that the ranking of high cartel bids is related to different determinants than the ranking of competitive or high bids by noncartel firms. Moreover, this paper finds no clear evidence that the probability of submitting bids differs between competitive and cartel bidders, as Porter and Zona (1999) found for school milk procurements by Cincinnati school districts in the 1980s. What this paper do find is a positive correlation between backlog and participation during the cartel period, suggesting that even if the cartel firms are (over)loaded with jobs, they tend to submit bids to create the impression of competition. Besides coordinating their bids, larger cartel members, in terms of the number of bids on different contracts each year, tended to bid less aggressively against each other during the cartel period. The pairwise test of conditional independence by Bajari and Ye (2003) indicates collusive bidding behavior among the convicted firms both during the cartel period and for some of the larger convicted firms after the court order in 2003. One potential explanation for this is that the testing procedure by Bajari and Ye does not discriminate between bids by the designated winner and phony bids by the other cartel members. Adding backlog, the spatial econometric approaches by Lundberg (2017) and Bergman *et al.* (2020) still verify collusive bidding behavior during the cartel period and reject such behavior for the postcartel period. This result is important as it suggests that the spatial econometric approach is

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robust to potential omitted variable bias. The result also provides more information to the findings of the bimodal distribution of bids by Aaltio *et al.* (2025); the phony or predetermined losing bids by members of the cartel are also correlated. That is, leaving a gap to the designated winning bid, the phony bids are not random but coordinated.

Practical implications – From the perspective of a competition authority, some considerations should be made before the authority initiates an investigation into a suspected cartel or conducts a more general scan of several different markets. Firstly, a competition authority often has limited resources and must therefore assess in most cases whether the monetary damage caused by the cartel is substantial enough to justify an investigation and the collection of further information. This should involve the direct economic damage that the cartel inflicts on the contracting authority, but also concern the extent to which the cartel affects the functioning of the market and thereby causes harm to competing companies and consumers. Secondly, are there sufficient procurements to enable an econometric analysis of the companies' bidding behavior? If not, further analysis must be based on other methods and the collection of additional information to substantiate any suspicions. Thirdly, if there is enough data for an econometric analysis, decide on which econometric method is best suited to detect the suspected behavior or for scanning.

Social implications – Coordinated bids on public contracts are probably more common than this paper realizes. As Harrington (2008) pointed out, *…cartels are among us. Their attempts to coordinate the prices they set and the quantities they produce are often effective. […] When we find them, we ought to prosecute and penalize them. But how do we find them?* Different econometric methods to detect coordinated bids have been proposed by, among others, Porter and Zona (1993, 1999), Bajari and Ye (2003), Lundberg (2017) and Bergman *et al.* (2020). But which one should we use? And do they reach the same conclusions? To what extent do they complement each other in our understanding of how the cartel operates?

Originality/value – This paper adds to the existing literature in at least three ways. First, this paper applies and evaluates the different methods on the same data. To our knowledge, this has not been done before with these specific methods. Such evaluation is important for national competition authorities in their choice of econometric method to detect and test for suspicious bid-rigging. Second, by introducing backlog in the econometric specifications used by Lundberg (2017) and Bergman *et al.* (2020), this paper tests the sensitivity and robustness of their results. Third, this paper complements recent and ongoing research on the evaluation of different econometric methods in this area (Huber and Imhof, 2019; Aaltio *et al.*, 2025). It is possible to obtain a bimodal distribution without the nonwinning cartel bids correlating with each other. It may also be the case that the participating companies rotate in different procurements, which does not create a correlation between the nonwinning bidders (as they do not participate in many procurements due to this rotation), while still achieving a bimodal distribution of the cartel's bids.

Keywords Auction, Procurement, Anti trust

Paper type Research paper

1. Introduction

Coordinated bids on public contracts are probably more common than we know. As pointed out by Harrington (2008), *Cartels are among us. Their attempts to coordinate the prices they set and the quantities they produce are often effective. […] When we find them, we ought to prosecute and penalize them. But how do we find them?* Different econometric methods to detect coordinated bids have been proposed by, among others, Porter and Zona (1993, 1999), Bajari and Ye (2003), Lundberg (2017) and Bergman *et al.* (2020) [1]. But which one to use? And do they come to the same conclusion? To what extent do they complement each other in our understanding of how the cartel operates?

In this paper we test five econometric methods for cartel detection suggested in the papers mentioned above on a data set that cover the so-called Swedish asphalt cartel where collusive bidding was detected by the Swedish Competition Authority (SCA) in 2001 and later verified by court in 2003. The contracts were awarded using open reverse first-price auctions. This data set contains information on procurements of paving and asphalt works (surfacing works) during the period 1995–2009 and the identity of convicted firms which makes it possible to analyze bidding behavior before and after the court order in 2003. The empirical strategy is to first test if the econometric methods suggested by Porter and Zona (1993, 1999) and Bajari and Ye (2003) are able to verify the asphalt cartel as did the methods suggested by Lundberg (2017) and Bergman *et al.* (2020). We also

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elaborate with the econometric specifications suggested by Lundberg and Bergman *et al*. The analysis concentrates on collusive bidding on the same contract. Finally, we also relate our findings to the analysis by Aaltio *et al.* (2025) who apply on the Swedish asphalt cartel the distributional regression approach formally stated and tested in Clark *et al.* (2025) on few known cartel cases and the machine learning approach formally stated and tested in Huber and Imhof (2019).

Detecting bid rigging from bid data is extremely difficult. Cartels are innovative and those who survive are obviously cleaver and innovative, perhaps more so than those chasing them. Collusive biding may take many forms and submitted bids could be designed to be consistent with competitive bidding. Regardless of the bidding strategy of the cartel, its members know that competition is limited and that they do not need to worry about being undercut by any other member of the cartel. Most papers on cartel detection concentrate on behavioral differences between cartel and noncartel bidders. The designated winner decided within the cartel are assumed to compete with potential bidders outside the cartel while the other cartel members will act in a noncompetitive manner and post bids most likely not related to their costs or backlog to create the appearance of competition. Hence, determinants of the lowest cartel bid (the designated winner) should not systematically differ from the other noncartel bids (who are supposed to act competitive) whereas the determinants of the competitive bids should differ from the complementary or phantom bids by the other cartel members. Hendricks and Porter (1989) empathize the importance to tailor empirical work to specific cases as competitive and collusive equilibria depend on the economic environment, tender documents and the nature of the good or service being procured. Moreover, based on various agreements within the cartel; cartel members can refrain from bidding on certain contracts, submit high or phantom bids on other contracts, and competitive bids on still other contracts. Therefore, it is unlikely that one single econometric method can detect and signal collusive bidding under a variety of conditions and circumstances. As pointed out by Porter and Zona (1993) and Bajari and Ye (2003), no econometric method or test of collusion is foolproof and will probably not without backup from other evidence convince any court or judge. They may however be used as a first step to scan markets for collusive behavior to indicate whether a further investigation is warranted. A legal process could be costly and a reliable econometric procedure to detect collusive behavior can save both time and public resources. It can also be used to backup and support other evidence in court and guide competition authorities in the functioning of the cartel and what type of other proof they should look for.

The data used and econometric approaches suggested by Porter and Zona (1993, 1999), Bajari and Ye (2003), Lundberg (2017) and Bergman *et al.* (2020) differ in some respects. Porter and Zona (1993, 1999) and Bajari and Ye (2003) only have (although strong) suspicions of collusive bidding in their data while Lundberg (2017) and Bergman *et al.* (2020) analyze bidding of a convicted cartel. Porter and Zona (1993) focus on the determinants of the levels of the submitted bids and the ordering of competitive and cartel bids. They argue that bids and rank distribution of noncartel firms should be related to cost while the bids and rank distribution of the higher cartel bids (all cartel bids except the designated winner who will compete against the noncartel firms) should not. Porter and Zona (1999) focus on two interrelated decisions: to either participate or not, and, if participating, what to bid. The basic hypothesis is that the probability of submitting a bid by competitive bidders should decrease with cost (in their case transportation costs) while this should not be the case for nondesignated winners within the cartel. And when submitting a bid, the bid placed by competitive and collusive bidders should differ systematically. Bajari and Ye (2003) state two necessary and sufficient conditions for a distribution of bids to be rationalized by their model of competitive bidding. One is conditional independence which means that after controlling for costs, bids must be independent. The other is exchangeability, which means that conditional on costs, the identities of competing firms

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should not affect the bid by any separate firm. They test both conditions on the bidding of two firms at a time. The test for exchangeability is also conducted on a constrained regression where the colluding firms are pooled together. In contrast to the other methods, Bajari and Ye incorporate the knowledge of industry experts into estimation and testing.

Lundberg (2017) and Bergman *et al.* (2020) use spatial econometric techniques to test if, conditional on other relevant covariates, bids by members of the cartel (but the lowest cartel bid) are correlated while bids by noncartel firms are not. This is in line with Porter and Zona (1993), who argue that the rank distribution of higher cartel bids should be more concentrated compared to noncartel bids, and the conditional independence condition by Bajari and Ye (2003). Both Lundberg and Bergman *et al*. apply their methods on the asphalt data used in this paper. While Lundberg apply the Moran's *I* test for spatial correlation both directly on the individual bids and on the residuals from a reduced form bid function, Bergman *et al*. estimate a spatial lag model [2]. In contrast to the other methods, Bergman *et al*. control for competition by including the number of bidders in the bid function.

The different econometric approaches presented above complement each other and applying them on the asphalt data gives additional insights into the functioning of the cartel. For instance, even though the ranking of the phantom bids by the nondesignated winners among cartel members do not systematically differ from the ranking of competitive bidders (Porter and Zona, 1993), they may still be correlated (Lundberg, 2017; Bergman *et al.*, 2020). Moreover, if the nondesignated winners among the cartel refrain from bidding (Porter and Zona, 1999), the methods suggested by Bajari and Ye, Lundberg and Bergman *et al*. will not detect conditional independence or correlation across the nondesignated winners as they do not bid. Recent research on cartel detection has focused on methods that relies on a minimum of information (or just the bids). The main motivation is that the collection of detailed cost-, project- and firm specific information is cumbersome and costly, and that competition authorities need methods to scan a large number of industries with a minimum of information [3]. One early paper in this line of research is Lundberg (2017) who apply a Moran's *I* test for correlation across losing bids. More recent work has applied other methods to flag for potential bid rigging such as low variance among cartel bids (Imhof *et al.*, 2018), isolating winning bids (Imhof *et al.*, 2018; Chassang *et al.*, 2022; Clark *et al.*, 2025), differences between close winning and losing bids (Kawai *et al.*, 2023) and machine learning (Huber and Imhof, 2019). Some of these recent studies are related to the methods applied in this paper. For instance, the study by Kawai *et al.* (2023) are related to Porter and Zona (1993). Both study the relationship between the rank order of bids and possible cost shifters to screen for collusion. While Porter and Zona (1993) focus on losing bidders, Kawai *et al.* focus on differences between winners and losers.

In a related paper, Aaltio *et al.* (2025) apply the distributional regression approach suggested by Clark *et al.* (2025) and the machine learning approach suggested by Huber and Imhof (2019) on the bidding behavior of the Swedish (the same data that is used in this paper) and the Finnish asphalt cartels that operated in the late 1990th and early 2000th. In both markets, they find that during the cartel period, a large share of the bids to be within 10-percent of the winning bid and winning bids to be isolated from the losing (or phantom) bids. The clustering of bids and the missing mass of nearly tied bids result in a bimodal distribution of the bids during the cartel period which is in line with the findings by Chassang *et al.* (2022), Clark *et al.* (2025) [4] and Imhof *et al.* (2018) [5]. Aaltio *et al.* (2025) also find the distributional regression approach to correctly reject competitive bidding for the cartel period in both the Swedish and the Finnish markets. However, the machine learning model suggested by Huber and Imhof (2019) provide mixed results. Depending on what training data used, the machine learning model did correctly classified tenders to be collusive or competitive in 70-percent to 87-percent of the tenders for Finland and 63-percent to

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74-percent of the tenders for Sweden. They note a substantial variation in the prediction rate of the machine learning methods to depend on the training data used.

In short, applied to the Swedish asphalt cartel and in contrast to the findings by Porter and Zona (1993) for the highway construction contracts on Long Island in the early 1980s, we don't find any systematic differences in bidding behavior between competitive and collusive bidders or evidence in favor of the hypothesis that the ranking of high cartel bids is related to other determinants than the ranking of competitive or high bids by noncartel firms. Moreover, we find no clear evidence that the probability to submit bids differs between competitive and cartel bidders as did Porter and Zona (1999) for school milk procurements by Cincinnati school districts in the 1980s. What we do find is a positive correlation between the backlog and participation during the cartel period which suggests that even if the cartel firms are (over) loaded with jobs, they tend to submit bids to make the impression of competition. Besides coordinating their bids, larger cartel members in terms of the number of bids on different contracts each year tended to bid less aggressive against each other during the cartel period. The pairwise test of conditional independence by Bajari and Ye (2003) indicates collusive bidding behavior among the convicted firms both during the cartel period and also for some of the larger convicted firms after the court order in 2003. One potential explanation for this is that the testing procedure by Bajari and Ye does not discriminate between bids by the designated winner and phony bids by the other cartel members. Adding backlog, the spatial econometric approaches by Lundberg (2017) and Bergman *et al.* (2020) still verify collusive bidding behavior during the cartel period and reject such behavior for the port-cartel period. This result is of importance as it suggests that the spatial econometric approach seems robust to potential omitted variable bias. The result also provides more information on the findings of the bimodal distribution of bids by Aaltio *et al.* (2025); the phony or predestinated losing bids by members of the cartel are also correlated. That is, leaving a gap to the designated winning bid, the phony bids are not random but coordinated.

This paper adds to the existing literature in at least three ways. First, we apply and evaluate the different methods on the same data. At least to our knowledge, this has not been done before with these specific methods. Such an evaluation is of importance to, for example, national competition authorities in their choice of econometric methods to detect and test for suspicious bid-rigging. Second, by introducing backlog in the econometric specifications used in Lundberg (2017) and Bergman *et al.* (2020), we test the sensitivity and robustness of their results. Third, this paper complements recent and ongoing research on the evaluation of different econometric methods in this area (Huber and Imhof, 2019; Aaltio *et al.*, 2025). It is possible to obtain a bimodal distribution without the nonwinning cartel bids correlating with each other. It may also be the case that the participating companies rotate in different procurements, which does not create a correlation between the nonwinning bidders (as they do not participate in many procurements due to this rotation), while still achieving a bimodal distribution of the cartel's bids.

The rest of this paper is organized as follows. The different methods are briefly described in section 2. A brief description of the procurement process in Sweden and the asphalt paving industry are given in section 3 followed by a presentation of our data set. The results from the different methods are presented in section 4 including the subsection 4.5 where the main results are summarized and put in perspectives with the findings by Aaltio *et al.* (2025). Concluding remarks are given in section 5.

2. Different methods for cartel detection

This section draws heavily on Porter and Zona (1993, 1999), Bajari and Ye (2003), Lundberg (2017) and Bergman *et al.* (2020).

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2.1 Porter and Zona *(1993)*

Porter and Zona (1993) propose a procedure where two models of bidding are estimated. They first estimate a bid function on three subsets of their data. One on all bids, one on competitive bids including the lowest cartel bid, and one on the other cartel bids. In the next step they estimate an ordered logit regression on the ranking of all bidders on these three subsets. In the absence of collusion, the parameter estimates from these estimations should not differ. If the lowest noncartel bidder's behavior is not statistically different from that of other noncartel firms while the determinants of the low cartel bid differ from those of higher cartel bids, this is taken as evidence for collusive behavior.

Porter and Zona apply their method on state highway construction contracts on Long Island in the early 1980s. Their data covers 116 projects, 186 contracts and 575 bids by 66 unique bidders. In 1984, one firm was convicted in federal court of rigging bids on one contract and four other firms listed as unindicted conspirators. All five firms have been named as participants in bid-rigging schemes in separate antitrust suites, each suit relates to one specific contract only and not to a general pattern. For the purpose of their paper, these five firms constitute the cartel. Joint bidding was allowed if the specific contract could only be performed using the combined capabilities of the participating firms. Porter and Zona take the fact that most firms submitted bids on most contracts as evidence against a bid rotation scheme. Instead, ring meetings prior to the auction served to designate a serious bidder while the others were expected to submit complementary or phantom bids. Product differentiation was not allowed and firms compete only in price.

To be more specific, the Porter and Zona estimation process goes as follows. Define by b_{it} the bid by firm i on contract t. Then estimate the log-linear bid-function:

$$ln(b_{it}) = \alpha_t + \beta X_{it} + e_{it} \tag{1}$$

by OLS where α_t is an auction-specific fixed effects, $\boldsymbol{X}_{it}$ is a vector of observable factors affecting the probability of firm i of winning or its observable costs on contract t, and e_{it} is private information. All observed auction specific information is assumed to be captured by the auction specific constant α_t[6]. The error term is assumed to have expectation zero and an auction-specific variance σ_t^2. Consistent estimates of the parameters hinge on the assumption that all eligible bidders are assumed to submit bids. A feasible generalized least square (GLS) is obtained by using the auction-specific variance from the OLS estimation to weight the data. Different bidding behavior for cartel versus noncartel firms would render in different parameter estimates for these two groups of firms.

In the next step, Porter and Zona focus on the ordering of phantom bids. Their reasoning goes as follows. By design, firms submitting phantom bids know that the designated winner will submit a lower bid than their own. Hence, phantom bids are only submitted to create the appearance of competition. However, the designated cartel winner must compete against bidders outside the cartel. Therefore, fundamental differences are likely to exist between the ordering of competitive and cartel bids conditional on the observed data. How the designated winner is chosen among the cartel firms is irrelevant. Unless the higher cartel bids are competitive, their ordering are not expected to reflect observable cost differences.

If based on costs, the behavior of cartel and noncartel firms should not differ. Porter and Zona concentrate on the ordering of the higher bids for each set of firms. Unless the higher cartel bids are competitive, their ordering should not reflect observable cost differences. Their data does not, however, contain any information on firm cost or contract specific characteristics. Instead, Porter and Zona base their test on behavioral differences between cartel and noncartel bidders.

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Denote the weighted, or deflated, data by $\boldsymbol{Z}$, the log probability that firm i will win the contract t is estimated as a standard multinomial logit:

$$lnP[b_{it} < b_{jt}, \forall j \neq i] = \alpha_t + \beta Z_{it} \quad (2)$$

where θ_t is an auction-specific constant, β and σ_t are coefficient from equation (1).

2.2 Porter and Zona (1999)

Porter and Zona (1999) focus on two interrelated decisions: where to bid and at what level. The main hypothesis is that collusive firms should act differently compared to competitive firms which boils down to a test of the parameter estimates based on competitive and collusive bidders. They estimate a reduced-form model of bidding behavior on a data set covering school milk procurements in 509 out of about 600 Cincinnati school districts between 1980 and 1990. They compare the bidding behavior of a group of firms prosecuted for bid-rigging in Cincinnati with a control group. They find the Cincinnati firms to bid on both local and distant contracts more often compared to firms in the control group. While nondefendant bids are an increasing function of distance, Cincinnati firms tend to post relatively low bids on distant contracts.

To be more specific, Porter and Zona divide their data set into two groups of bidders. Those who have been accused of collusive behavior and those who have not. The probability to participate are then estimated for both groups. In the absence of collusion, the parameter estimates for the two groups should not differ. Next, Porter and Zona test for differences between what the separate collusive bidders should have bid and their actual bid. Their testing strategy goes as follows. First, they compute the predicted bids for each one of the collusive bidders based on the bid equation of noncollusive bidders only. This will predict what the individual collusive bidders would have bid if they did not collude. Second, the difference between the predicted and the actual bid is calculated and, third, regressed on the right-hand side variables in the bid equation. Finally, a*n F*-statistic is computed to test if all coefficients are equal to zero. A rejection of the null hypothesis indicates that collusive firms bid differently compared to noncollusive bidders. Porter and Zona also test for statistical independence in the probability of bidding on the same contract and on the residuals from the reduced bid *equation.*

2.3 Bajari and Ye (2003)

Bajari and Ye (2003) state a set of both necessary and sufficient conditions for a distribution of bids to be generated by a model of competitive bidding. These conditions are conditional independence and exchangeability. Conditional independence means that after controlling for all information about costs that are publicly observed by the bidders, the bids should be independent. In a situation with collusion, correlation between cartel members' bids is expected when cartel members post "phantom" bids. Exchangeability implies that costs alone should determine how firms bid and not be affected by the identity of competing firms. Cartel members are not expected to bid as aggressively against each other as a control group of noncartel firms. Conditional independence and exchangeability are tested for two companies at a time and not for a group of companies.

One unique component in their analysis is that the beliefs of industry experts are incorporated in the model parameters, estimation, and testing. Their model also allows for asymmetric bidders which, in a procurement setting, relax the assumption that private cost estimates are independently and identically distributed across firms. Such asymmetries can

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easily arise from different locations, different capacity constraints, different technologies, and different levels of managerial efficiencies across firms.

Without proof or prior knowledge of collusive behavior, conditional dependence and exchangeability are tested for on the seal coat industry in the US. In total, their data set covers bids by 98 different companies on at least one of 495 public contracts in 138 procurements in Minnesota, North and South Dakota between January 1994 and October 1998 [6].

To test for conditional independence and exchangeability, Bajari and Ye first estimate:

$$\frac{b_{it}}{EST_t} = \beta_{0k} + \beta_{1k} LDIST_{it} + \beta_{2k} CAP_{it} + \beta_{3k} MAXP_{-it} + \beta_{4k} LMDIST_{-it} + \beta_{5k} CON_{it} + \omega_i + \delta_t + \epsilon_{it} \quad (3)$$

where b_{it} is the bid by firm i on contract t, *EST* the estimated costs by industry experts, $LDIST = log(DIST_{it} + 1)$ with *DIST* defined as the distance between the location of the firm and the project which is supposed to reflect firm specific costs, *CAP* is a capacity measure, *MAXP* and *MDIST* reflect maximum percentage free capacity of competitors and the minimum distance of competitors to the site respectively, and *CON* is the proportion of work done (by dollar volume) by the firm.*LDIST, CAP* and *CON* affect firm cost while *MAXP* and *LMDIST* reflect cost of competing firms. ω_i and δ_t are firm (only for the largest firms) and auction fixed effects. *EST* were only available for 139 of the 441 projects in the data set.

To make the model flexible, separate parameters are estimated for each of the cartel firms while noncartel firms are assumed to have the same parameters. Let c be the correlation coefficient between ϵ_{it} and ϵ_{jt}, the test of conditional independence between firm i and j are then based on the statistic:

$$z = \frac{1}{2} ln\left(\frac{1+c}{1-c}\right)\sqrt{n-3} \quad (4)$$

calculated for $n > 3$ where n is the number of times bidder i and j bid on the same contract.
To test for exchangeability, which implies that capacities and distances should enter the firm's bid value in a symmetric way, they test the null-hypothesis:

$$H0 : \beta_{lj} = \beta_{li} \text{ for all } i, j, i \neq j \text{ and } l = 1, \ldots, 4 \quad (5)$$

The test for exchangeability is conducted both on the constrained regression of equation (3) where the colluding firms are pooled together and on a pairwise basis where two of the cartel firms are pooled together.

2.4 Lundberg (2017) and Bergman et al. (2020)

Both Lundberg (2017) and Bergman *et al.* (2020) build on the spatial econometric techniques frequently used to test for potential interaction effects across local governments, regions or states [7]. Both studies apply their approach on the so-called Swedish Asphalt cartel detected in an unannounced morning raid by the Swedish Competition Authority (SCA) in October 2001 described above. Based on this information and a data set covering asphalt paving before and after the raids, the methods proposed by both Lundberg and Bergman *et al.* can verify collusive behavior among convicted firms before 2001 while no such behavior can be verified after 2003.

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The basic idea is that a dependence should exist across the higher cartel bids while not across the competitive bidders. Consider two types of bidders, *X* and *Y*, who compete for a contract awarded the lowest bidder. Bidders of type *X* form a cartel while bidders of type *Y* act independently. Denote by b_{it} a bid placed by bidder *i* on contract *t*, n_t the total number of contracts, n_X and n_Y the average number of type *X* and *Y* bidders on each contract and the total number of bids on all contracts $n = n_t(n_X + n_Y)$. Type *X* bidders decide on a designated winner who place a competitive bid, bidder *k*, while the other type *X* bidders (*j*, *l*) engage in complementary bidding and place high bids. Construct a $(N \times N)$ matrix $\boldsymbol{W}$ with elements defined as $w_{jl,\ lt} > 0$, else $w = 0$. Next $\boldsymbol{W}$ is row-standardized such that each row with at least one element $w > 0$ summarize to 1 [8]. To test for collusion, Bergman *et al.* (2020) estimate:

$$b_{it} = \alpha + X\beta + \rho W b_{it} + \varepsilon_{it} \tag{6}$$

where *b* is a vector of all bids placed by both types of bidders, $\boldsymbol{X}$ contain all relevant covariates, ε is an error term with the usual properties and α, β and ρ are parameters to be estimated. Given $\boldsymbol{W}$, this implies that bidding strategies on a contract *t* are independent across type *Y* bidders, independent between type *X* and type *Y* bidders while not across the type *X* bidders who place complementary bids. Hence, if $\rho \neq 0$ a bid placed by type *X* bidder *j* depend on the bid placed by type *X* bidder *l* and vice versa [9]. If $\boldsymbol{X}$ do not cover all relevant covariates, $\rho_{cartel} \neq \rho_{non-cartel}$ is expected. Bergman *et al.* (2020) also estimate a log version of equation (6).

Lundberg (2017) use the Moran's *I* test for spatial correlation based on the same definition of the elements in the weight's matrix $\boldsymbol{W}$ as in Bergman *et al.* (2020). The Moran's *I* statistic can either be based directly on the bids or on the residuals from a regression like equation (6) with the restriction $\rho = 0$. Applying the Moran's *I* directly on the bids, the Moran's *I* is calculated as:

$$I = \left(\frac{N}{s_0}\right)\frac{b'Wb}{b'b} \tag{7}$$

or on the residuals:

$$I = \left(\frac{N}{S_0}\right)\frac{\varepsilon'W\varepsilon}{\varepsilon'\varepsilon} \tag{8}$$

where $S_0 = \sum_i \sum_j w_{ij}$ which is equal to 1 if $\boldsymbol{W}$ is row standardized. The test statistic is compared to its theoretical mean $E(I) = -1/(n-1) \rightarrow 0$ as $n \rightarrow \infty$. The null hypothesis of no dependence across bids by type *A* bidders, H_0: $I = -1/(n-1)$, is tested against the alternative, H_a: $I \neq -1/(n-1)$. Inference is based on $z = (I - E(I))/SD(I)$ where $SD(I)$ is the theoretical standard deviation of *I*. If H_0 is rejected there are two alternative interpretations depending on $I > 1/(n-1)$ or $I < 1/(n-1)$. $I > -1/(n-1)$ indicate that type *A* firms, except the designated winner among type *A* bidders, place more similar bids on a specific contract than would be expected by pure chance which is consistent with complementary bidding. On the other hand, if $I < -1/(n-1)$ type *A* firms put a mixture of high and low bids on the same contract.

The Moran's *I* presented in Lundberg (2017) are calculated on the pure bids (b_{it}), $ln(b_{it})$ and the residuals from different versions of the OLS regression:

$$b_{it} = \alpha + \beta_{comp} \times comp_t + \beta_{dens} \times dens_r + \beta_{area} \times area_t + \alpha_p + \alpha_t + \alpha_r + \alpha_i + \varepsilon_{itr} \tag{9}$$

where the α's are year, contract, region, and firm fixed effects.

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The spatial lag model is related to the Bajari and Ye test for conditional dependence and the Moran's *I* test in the bids is a test for unconditional independence across bidders. The main difference is that Lundberg and Bergman *et al*. both distinguish between high and low collusive bidders while Bajari and Ye do not.

3. Public procurement in Sweden and data

3.1 Public procurement in Sweden

In Sweden as in many other OECD countries, competitive sealed bidding is generally used to allocate public contracts. The auction is announced through a call for tender where, among other things, auction rules, technical characteristics of the goods and/or services to be procured and supplier selection are specified, see Dini *et al*. (2006), Asker and Cantillon (2008, 2010) and Bergman and Lundberg (2013) for a in depth description of the process. Several contracts may be auctioned in one and the same procurement. Bidding is simultaneous but independent over contracts, meaning that separate bids are submitted on all or a subset of the contracts included in the same procurement. Each potential supplier is allowed to submit only one bid per contract. In general, the winning bidder is paid in accordance with the price offered in the tender. If the total value of the procurement is below the threshold value [10], the procurer can use a simplified procedure that is open for all bidders, and the procuring authorities have the opportunity to initiate negotiations with some of the bidders. For procurements above the threshold, the open procedure is open to everyone, while the procedures negotiated and restricted have some limitations in bidding, e.g. firms must qualify to be given the opportunity to bid. For purchases of relatively low value, authorities are allowed to use a less formal auction format or a so-called direct procurement. Procurements above the threshold value must be announced in the Official Journal of the European Union.

3.2 Asphalt paving industry and the asphalt dataset

The Swedish Road Administration (SRA) is responsible for construction, operation and maintenance of national roads making SRA a large and frequent byer of road pavement and repair. Most calls for tender for road pavement contracts are posted early in the spring and most of the work is carried out during the summer. The bidders are allowed to use each other as subcontractors, which makes capacity constraints less of a concern using this data. The contracts were awarded using open reverse first-price auctions.

In late October 2001, the Swedish Competition Authority (SCA) (2007, 2009) conducted unannounced raids on a number of companies within this industry with the purpose to find documents to support their suspicions of illegal collusive bidding on public contracts. Later, in 2003, six [11] separate firms were convicted of collusive bidding in the Stockholm District Court. Although there were suspicions that cartel cooperation took place throughout the country and involved all firms (even small local firms), the conviction only applied to procurements of asphalt work in southern Sweden.

The data set used in this paper covers 232 procurements of 427 contracts of paving and asphalt works (surfacing works) over the period 1995–2009 and originate from three sources: the Swedish Road Administration (SRA); Statistics Sweden (SCB) and the Swedish Meteorological and Hydrological Institute (SMHI). The data set contains information on a total of 2,129 bids submitted by 58 different firms. The convicted firms are treated as collusive firms. The conviction makes it possible to test for different bidding behavior before and after the morning raids in 2001 and court order in 2003 [12]. Descriptive statistics are presented in Table 1.

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Table 1. Descriptive statistics

Variables	Mean	SD	Min.	Max.
Whole sample, 1995–2009				
Bid, SEK per square meter (b_{it})	1,630.6	7,685.1	6.19	92,161.5
Bid / winning bid (rwb_t)	1.123	0.461	0.110	11.843
Square meters per contract ($area_t$)	55,386	71,701	160	607,613
Backlog ($blog_{it}$)	0.402	0.468	0	1.928
Competition; No of bids ($bids_t$)	5.42	1.50	2	10
Potential number of bidders ($pbids_t$)	107.42	30.95	68	192
Population density, pop per km^2 ($dens_r$)	64.55	62.43	3.29	196.9
Average temperature, Celsius ($temp_t$)	6.00	2.31	0.35	9.68
Average rainfall, millimeter ($rain_t$)	56.41	12.65	29.43	81.89
Number of contracts	427			
Observations	2,129			
1995–2001 (cartel period)				
Bid, SEK per square meter (b_{it})	1,717.2	8,036.8	12.1	92,161.5
Bid/winning bid (rwb_t)	1.116	0.386	0.110	11.843
Square meters per contract ($area_t$)	47,937	61,138	160	607,613
Backlog ($blog_{it}$)	0.428	0.491	0	1.928
Competition; No of bids ($bids_t$)	5.61	1.47	2	10
Potential number of bidders ($pbids_t$)	108.68	31.49	68	167
Population density, pop per km^2 ($dens_r$)	72.28	63.47	3.30	196.91
Average temperature, Celsius ($temp_t$)	6.39	1.96	0.36	9.65
Average rainfall, millimeter ($rain_t$)	56.31	13.28	29.43	81.89
Number of contracts	352			
Observations	1,830			
2004–2009 (postcartel period)				
Bid, SEK per square meter (b_{it})	1,100.4	4,999.5	6.19	39,300
Bid / winning bid (rwb_t)	1.170	0.789	0.803	10.664
Square meters per contract ($area_t$)	100,978	106,513	170	601,625
Backlog ($blog_{it}$)	0.247	0.239	0	0.850
Competition; No of bids ($bids_t$)	4.26	1.06	2	7
Potential number of bidders ($pbids_t$)	99.71	26.22	76	192
Population density, pop per km^2 ($dens_r$)	17.26	22.15	3.29	108.76
Average temperature, Celsius ($temp_t$)	3.63	2.85	0.67	9.68
Average rainfall, millimeter ($rain_t$)	57.03	7.68	45.32	75.72
Number of contracts	75			
Observations	299			

Note(s): Bid per square meter is measured in Swedish Kronor (SEK) per square meter in the 2009 price level; The Swedish asphalt cartel
Source(s): Created by author

The separate bids by a firm *i* on contract *t*, b_{it}, is measured as the bid in SEK per square meter of paving ($area_t$). Backlog ($blog_{it}$) is defined as the number of contracts won by firm *i* times the average area of these contracts during the spring when most calls for tender for road pavement contracts are posted. The variable $bids_t$ is defined as the actual number of bidders on each contract and the potential number of bidders $pbids_t$ is the number of existing firms in the region where the contract is issued. Population density ($dens_r$), the average temperature ($temp_r$), and average rainfall ($rain_t$) are defined as the population density, average yearly temperature and average yearly rainfall in regio*n r*. These variables are used to capture potential differences in transportation costs ($dens_r$) and regional conditions ($temp_r$ and $rain_r$).

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It is interesting to note that during the cartel period, the convicted firms won 275 out of 427 contracts (64-percent) and 54 out of 75 (72-percent) in the period after 2003. Moreover, the relation between the individual bids and the winning bid (rwb_{it}) is lower for the later period, which is in line with a more competitive bidding during that period.

Note that the standard deviation of the bids, *b*, decreased during the postcartel period. This observation aligns with the findings reported by Imhof *et al.* (2018). In contrast, both Feinstein *et al.* (1985) and Abrantes-Metz *et al.* (2006) observed that the variation is lower when firms engage in collusion. Athey *et al.* (2004) and Harrington and Chen (2006) propose two distinct explanations for why the variability in bids should be reduced under a collusive arrangement. Athey *et al.* consider an infinitely repeated Bertrand game in which, during each period, firms exchange information regarding their costs before deciding on prices (or bids). However, individual firms lack the incentive to disclose their true costs. Athey *et al.* demonstrate that if firms are sufficiently patient, an optimal collusion will result in rigid bids. Harrington and Chen argue that cartels, in an effort to avoid detection, will be less responsive to cost shocks, thereby reducing the variability in bids. Without deeper analysis, it is only possible to speculate on the reasons for the increase in standard deviation as reported in Table 1.

The number of bids by cartel firms on the same contract are shown in Table 2 and the number of bids in Table 3 [13]. Firms *A*, *E* and *F* are the most frequent bidders and often bid on the same contract.

4. Empirical cartel detection

Before we proceed and confront the different methods with our data, it might be justified to comment on functional form, specifications, omitted variables, inclusion of different fixed effects, etc. Our point of departure is the functional forms used in the original papers. There are of course other functional forms, specifications, etc. but to save space and limit our analysis we make this choice. Here, we include firm, regional, and year fixed effects. This decision is informed by previous studies on cartel detection using the same data set, such as Lundberg (2017) and Bergman *et al.* (2020), as well as court orders from Marknadsdomstolens dom (2004, 2009) and documentation from the Swedish Competition Authority (SCA) (2007). Firm fixed effects are incorporated to account for firm-specific unobservable factors, such as costs. Regional fixed effects are included to capture unobserved regional characteristics that might influence the formation of collusive behavior. Year fixed effects are used to control for common macroeconomic shocks. Contract-specific characteristics are controlled for by including the size of the project (*area*) measured in square meters. Contract fixed effects are excluded as they explain 99% of the variation in the dependent variable b_{it} [14].

Table 2. Number of bids on the same contract during the cartel period (after the cartel period) in the upper part and bid correlation during (after) the cartel period in the lower part

	A	B	C	D	E	F
A	–	46 (2)	158 (23)	86 (12)	322 (41)	302 (52)
B	1.000 (.)	–	40 (1)	8 (1)	47 (1)	47 (2)
C	0.999 (0.984)	0.999 (.)	–	33 (6)	159 (19)	152 (19)
D	1.000 (1.000)	0.983 (.)	0.994 (0.991)	–	87 (7)	62 (12)
E	0.999 (0.999)	0.999 (.)	0.997 (0.988)	0.999 (0.998)	–	301 (32)
F	0.999 (0.960)	0.999 (.)	1.000 (0.977)	0.996 (0.999)	0.997 (0.936)	–

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Table 3. Bidding activities, cartel firms

Firm ID	Bids in sample	No. of wins	Av. relative bid
1995–2001			
A	331 (94%)	79 (24%)	1.031
B	47 (13%)	5 (11%)	1.009
C	162 (46%)	21 (13%)	1.006
D	87 (25%)	27 (31%)	1.016
E	339 (96%)	82 (24%)	1.030
F	312 (87%)	61 (20%)	1.031
2004–2009			
A	66 (88%)	17 (26%)	1.006
B	2 (3%)	0 (0%)	1.000
C	25 (33%)	7 (28%)	1.000
D	13 (17%)	2 (15%)	1,000
E	44 (59%)	10 (23%)	1.007
F	61 (81%)	18 (30%)	1.006

Source(s): Created by author

Models using *rwb* as the dependent variable are estimated for comparison with the Bajari and Ye paper where the bids are normalized by the estimated cost by industry experts. To limit the range of the parameter estimates, the variables are scaled differently in different specifications. The covariates are restricted to the available information on contract, firm, and regional characteristics.

It should also be noted that even though not convicted, we have no waterproof proofs but only strong reasons to assume that the asphalt cartel firms did not coordinate their bids during what we refer to as the post cartel period.

4.1 Porter and Zona *(1993)*

The basic idea behind the Porter and Zona approach is that as the designated winner competes with the other noncartel firms, the designated winner should act as a competitive firm along with the other noncartel firms. Hence, the bidding strategy by the designated winner and the other noncartel firms should in a systematic way depend on costs and other relevant characteristics. On the other hand, the high cartel bids are phony bids and should therefore not be related to the same characteristics [15]. This applies for both the bids and the ranking of the bids. The reduced bid equation is estimated for three subgroups; all firms, competitive firms including the lowest cartel bid, and high cartel bids. The ranking is estimated on all rankings, high cartel rankings (all cartel firms except the lowest cartel bidder), high noncartel rankings (all noncartel firms except the lowest bidder), and all noncartel bidders. A systematic difference between the determinants of the high cartel bids and the competitive bids indicates collusive bidding behavior.

Estimations based on the strategy suggested by Porter and Zona are presented in Tables 4 and 5. The covariates in the GLS and MNL estimations are weighted by the contract specific variance from a first stage OLS regression with the same set of covariates and firm, contract, and year fixed effects. The GLS model is estimated using the same contract characteristics as Lundberg (2017) and Bergman *et al.* (2020) but without contract fixed effects and competition. To capture backlog, we include the number of bids per year and the size of contracts won during that year as a firm specific characteristic, $blog_{it}$. The MNL model does not converge using firm fixed effects.

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Table 4. GLS estimates; dependent variable *ln*(*b*)

	Cartel period 1995–2001			Post cartel period 2003–2009		
Variables	All firms	Competitive	High cartel	All firms	Competitive	High cartel
Area (β_{area})	−0.949*** (0.08)	−1.149*** (0.13)	−0.791*** (0.11)	−2.424*** (0.44)	−3.818*** (0.90)	−1.896*** (0.47)
Pop density (β_{dens})	−0.483*** (0.05)	−0.601*** (0.07)	−0.028 (0.17)	9.883** (4.88)	2.955 (9.20)	11.271** (5.59)
Temperature (β_{temp})	2.052*** (0.21)	2.342*** (0.32)	1.427*** (0.40)	1.896 (2.57)	3.013 (4.52)	−0.406 (3.10)
Rainfall (β_{rain})	−0.499*** (0.17)	−0.433 (0.28)	−0.321 (0.23)	−0.911 (0.92)	2.656 (2.42)	−0.816 (0.99)
Bids (β_{bids})	0.008** (0.00)	0.001 (0.01)	0.031*** (0.01)	0.028 (0.04)	0.036 (0.07)	−0.001 (0.06)
Backlog (β_{blog})	−0.005 (0.01)	−0.003 (0.01)	−0.047 (0.07)	0.274 (1.84)	−5.757 (4.52)	0.657 (2.16)
Fixed effects						
Firm	Y	Y	Y	Y	Y	Y
Regional	Y	Y	Y	Y	Y	Y
Year	Y	Y	Y	Y	Y	Y
Observations	1,830	902	928	299	163	136
R^2_{adj}	0.208	0.235	0.211	0.262	0.221	0.272
AIC	6,141	2,902	3,198	1,017	603	427

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Table 5. Multinomial logit (MNL) estimates

Variables	Cartel period 1995–2001				Postcartel period 2004–2009			
	All firms	High cartel	High noncartel	Comp	All firms	High cartel	High noncartel	Comp
Area (β_{area})	0.143 (0.11)	0.083 (0.16)	0.248 (0.80)	0.084 (0.49)	−0.891 (0.60)	−0.352 (0.84)	−45.29 (42.53)	−1.340 (4.70)
Pop dens (β_{dens})	0.062 (0.07)	−0.194 (0.23)	−0.166 (1.08)	0.121 (0.30)	12.280** (6.60)	−0.733 (9.83)	−1.09e + 04 (20644)	74.55 (57.40)
Temp (β_{temp})	−0.090 (0.31)	−0.280 (0.58)	0.216 (2.22)	−0.712 (1.14)	0.257 (3.42)	−6.090 (5.39)	890.3 (1604)	−2.138 (20.71)
Rain (β_{rain})	−0.103 (0.23)	0.042 (0.32)	−1.115 (2.21)	0.368 (1.25)	−1.540 (1.24)	0.529 (1.73)	80.16 (167.6)	−9.870 (8.55)
Bids (β_{bids})	−0.019*** (0.00)	−0.025*** (0.00)	−0.040*** (0.01)	−0.035*** (0.01)	−0.046 (0.03)	−0.143** (0.06)	−0.145 (0.28)	−0.267** (0.13)
Backlog (β_{blog})	−0.078 (0.09)	0.231 (0.21)	−2.953 (2.23)	−1.768** (0.88)	−2.659 (2.52)	7.266 (3.73)	31.07 (114)	−24.05 (39.02)
Fixed effects								
Firm	N	N	N	N	N	N	N	N
Regional	Y	Y	Y	Y	Y	Y	Y	Y
Year	Y	Y	Y	Y	Y	Y	Y	Y
Observations	1,830	928	291	552	299	136	37	88
AIC	6861	2324	620	1303	997	300	71	182

Note(s): Standard deviation within parenthesis. *$p < 0.10$, **$p < 0.05$, ***$p < 0.01$; Bid rank data
Source(s): Created by author

The GLS estimates of bid levels are displayed in Table 4. First, the explanatory power is low for all specifications. The parameter estimates are more precisely estimated for the cartel period and also more related to the cost measures used here. Based on the standard deviations, there is no systematic difference in bidding behavior between competitive and collusive bidders in neither period. In short, from the GLS estimates presented in Table 4 provides no (strong) evidence that the bidding behavior during the cartel period differs between competitive bidders and the nondesignated winners among the cartel firms.

The MNL estimates of bid ranking based on the same covariates as in the GLS estimations are displayed in Table 5 [16]. To verify the cartel, the ranking of competitive bids and cartel bids should be systematically different during the cartel period while not during the noncartel period. Among the cartel firms, the backlog tends to have a negative effect on the ranking which suggests that the more contracts won by a cartel firm, the lower the ranking. Even if not significant, this behavior is also present among the high noncartel bidders. More interesting is the fact that, based on the estimates presented in Table 5, there is no evidence in favor of the hypothesis that the ranking of high cartel bids is related to other determinants than the ranking of competitive or high noncartel bids.

Turning to the post cartel period, our data contains only 37 high noncartel bids and the parameters are estimated with very high standard deviations. Therefore, we concentrate on comparison of the competitive and high cartel bids. Again, there is no clear evidence that the ranking of high cartel bids systematically differs from the ranking of the competitive bids, which is in line with expectations for this period.

The results presented in Table 4 and 5 do not give any strong indications of collusive bidding behavior among the convicted firms. The only result that points in that direction is the fact that population density is estimated to have a negative impact on the bids during the cartel period while a positive effect during the post cartel period. There are, however, no clear differences between the behavior of the convicted firms and the control group. One potential explanation for the fact that the results based on Porter and Zona (1993) do not provide clear evidence for collusive bidding on these contracts is the lack of more detailed firm and cost characteristics. However, the results do correctly reject the hypothesis of collusive behavior for the postcartel period.

4.2 Porter and Zona (1999)

Porter and Zona (1999) concentrate on two decisions. First the decision to participate and, given that the firm participate, compare the bids of competitive and collusive bidders. The main hypothesis is that collusive firms should act differently compared to competitive firms which boils down to a test of the parameter estimates based on competitive and collusive bidders. Parameter estimates are based on three subgroups of our data set: all firms, noncollusive firms (competitive) and collusive firms. The hypothesis is that the decision to participate by collusive firms should be determined by other factors compared to the same decision by noncollusive firms and that collusive firms should participate more often just to give the impression of competition. The results are presented in Table 6.

The results displayed in Table 6 show no clear evidence that the probability to submit bids differ between competitive and cartel bidders nor between the two periods. Surprisingly, the backlog is estimated to have a positive effect on participation for all subgroups and periods, even though the effect is not significant for competitive and collusive bidders during the post cartel period. One potential explanation for the positive effect on the participation of cartel firms during the cartel period is that, even if the cartel firms are loaded with jobs, they tend to post (phony) bids to make the impression of competition. However, this positive effect is also

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Table 6. Parameter estimates participating

Variables	Cartel period 1995–2001			Post cartel period 2004–2009		
	All firms	Competitive	Collusive	All firms	Competitive	Collusive
Pop density (β_{dens})	−0.003 (0.03)	0.013 (0.02)	−0.030 (0.03)	−0.370 (0.36)	−0.758 (0.66)	−0.568 (0.53)
Area (β_{area})	0.322 (0.40)	0.462 (0.50)	0.212 (0.75)	−0.499 (0.55)	−0.189 (0.91)	−1.178 (0.82)
Temperature (β_{temp})	−1.174 (1.65)	−2.032 (2.05)	0.435 (2.67)	1.770 (3.38)	6.015 (4.72)	−5.547 (5.64)
Rainfall (β_{rain})	−0.121 (0.13)	−0.187 (0.17)	0.019 (0.23)	−0.235 (0.26)	−0.511 (0.61)	−0.580 (0.40)
Bids (β_{bids})	−0.003 (0.01)	0.001 (0.01)	−0.011 (0.01)	−0.101*** (0.04)	−0.137** (0.05)	−0.060 (0.06)
Backlog (β_{blog})	0.370*** (0.10)	0.566* (0.32)	0.377*** (0.12)	0.665* (0.40)	1.511 (1.35)	0.381 (0.47)
Fixed effects						
Firm	Y	Y	Y	Y	Y	Y
Regional	Y	Y	Y	Y	Y	Y
Year	Y	Y	Y	Y	Y	Y
Observations	6,902	5,015	1,887	868	480	388
AIC	4,057	2,570	1,438	853	420	378

Note(s): Standard deviation within parenthesis. *$p < 0.10$, **$p < 0.05$, ***$p < 0.01$
Source(s): Created by author

present for the post cartel period and does not significantly differ between the periods, even though it's not statistically significant for the later period.

Table 7 displays chi-2 test of the hypothesis $\beta_{\cdot}^{cartel} = \beta_{\cdot}^{non-cartel}$ for the two periods. Based on the chi-2 tests we cannot reject the hypothesis of equal parameter estimates for the two subgroups of competitive and cartel firms.

Next, Porter and Zona test for differences between what the separate collusive bidders should have bid and their actual bid. Their testing strategy goes as follows. First, they compute the predicted bids for each one of the collusive bidders based on the bid equation of noncollusive bidders only. This will predict what the individual collusive bidders would have bid if they did not collude. Second, the difference between the predicted and the actual bid is calculated and, third, regressed on the right-hand side variables in the bid equation. Finally, a*n F*-statistic is computed to test if all coefficients are equal to zero. A rejection of the null hypothesis indicates that collusive firms bid differently compared to noncollusive bidders. The *F*-statistics for the convicted firms for the two periods are displayed in Table 8 below.

For the cartel period, the null hypothesis is rejected at the 5-percent level for firms *A*, *E* and *F* and at the 10-percent level for firm *C*. Hence, the null hypothesis of competitive bidding behavior is rejected at the 10-percent level for 4 out of 6 convicted firms. Looking at bidding frequencies (see Table 3) *A*, *C*, *E* and *F* are the most frequent bidders. One potential explanation for the result is that those four dominating firms dictate the cartel and when bidding, they are able to lift prices. When *B* and *D*, who are least frequent bidders, participate, and are assigned to be the designated winner, they do not have the same market power as the larger cartel firms and therefore feel the need to act in a more competitive manner to win the contract.

Table 7. Parameter tests, chi-2

Variables	Cartel period 1995–2001	Post cartel period 2004–2009
Pop density (β_{dens})	1.60 ($p = 0.205$)	0.05 ($p = 0.823$)
Area (β_{area})	0.08 ($p = 0.781$)	0.66 ($p = 0.418$)
Temperature (β_{temp})	0.54 ($p = 0.464$)	2.47 ($p = 0.116$)
Rainfall (β_{rain})	0.52 ($p = 0.470$)	0.01 ($p = 0.924$)
Bids (β_{bids})	1.34 ($p = 0.247$)	0.94 ($p = 0.331$)
Backlog (β_{blog})	0.31 ($p = 0.580$)	0.63 ($p = 0.428$)
All	386.49 ($p = 0.000$)	95.38 ($p = 0.000$)

Note(s): *p*-values within parenthesis
Source(s): Created by author

Table 8. *F*-statistics and *p*-values

Firm	Cartel period 1995–2001	Post cartel period 2004–2009
Firm A	2.20 ($p = 0.005$, df = 298)	∞ ($p = 0.000$, df = 37)
Firm B	0.63 ($p = 0.792$, df = 24)	(No observations)
Firm C	1.77 ($p = 0.052$, df = 135)	(No degrees of freedom)
Firm D	1.05 ($p = 0.417$, df = 58)	(No degrees of freedom)
Firm E	2.06 ($p = 0.010$, df = 306)	∞ ($p = 0.000$, df = 15)
Firm F	2.41 ($p = 0.002$, df = 279)	∞ ($p = 0.000$, df = 34)

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Turning to the post cartel period, the null hypothesis of competitive bidding is rejected for cartel firms *A*, *E* and *F*. This suggests that the bidding behavior of the convicted firms where not in accordance with a competitive bidding during the later period. One potential explanation is the low degrees of freedom making the test less reliable.

To summarize, there are no clear indications that the decision to participate differs across the cartel and noncartel firms. However, the larger cartel firms tend to bid in a less competitive manner during the cartel period compared to the noncartel firms which are in line with collusive bidding.

4.3 Bajari and Ye (2003)

Bajari and Ye (2003) provide formal tests for conditional independence and exchangeability. They normalize the bids by the estimated cost by industry experts to carry out the contract. The average winning bid in their paper is 1.0031 with a standard deviation of 0.1573 which indicates that the winning bid is often close to the predicted cost by industry experts. Information from industry experts is not available to us. Instead all bids are normalized by the winning bid, denoted *rwb*, which is not always the same as the lowest bid [17]. For comparison, all tests and estimations are done using the alternative measures bids per square meters (*b*) and *ln*(*b*).

The results of the reduced bid function presented in Table 9 give no clear indications of a different bidding behavior during the cartel period compared with the post cartel period. One potential explanation is of course the mix of competitive and collusive bidders during the cartel period.

The formal test for conditional independence and exchangeability is presented in Table 10. The Fisher *z* test for conditional independence during the cartel period (post cartel period) are presented in the upper part of the table and *F*-test probabilities for exchangeability for the cartel period (post cartel period) in the lower part. For the cartel period and using the same definition as Bajari and Ye (*rwb*), the success rate for their test is 7 out of 15 (or 50-percent), 13 out of 15 using *b* (87-percent) and 14 out of 15 using *ln*(*b*) (93-percent). The same numbers for the postcartel period are 90-percent, 70-percent and 10-percent respectively. The main conclusion is that using relative bids is the most conservative approach while the log specification tends to overestimate the presence of conditional independence. This success rate can be compared to the success rate of the machine learning model presented in Aaltio *et al.* (2025) which did correctly classified tenders to be collusive or competitive in between 63-percent to 74-percent of the tenders for Sweden.

The pairwise test procedure for conditional independence suggests that bids by cartel members are not independent which is in line with the findings by Lundberg (2017) and Bergman *et al.* (2020). One difference is that Bajari and Ye do not distinguish between the designated winner and the other higher cartel bids. This might be one potential explanation why some combinations of cartel members are not detected. However, the rejection of conditional independence between firms *A* and *C*, *A* and *E*, *A* and *F*, *C* and *E*, *C* and *F*, and firms *E* and *F* is in line with the rejection of competitive bidding by firms *A*, *C*, *E* and *F* using the strategy by Porter and Zona (1999).

Turning to exchangeability and the probabilities from the *F*-test shown in the lower part of Table 10. During the cartel period and based on *rwb*, only one pair of firms (*B*, *F*) fail the test at the conventional 5-percent level of confidence and only two pairs (*B*, *C* and *B*, *E*) fail the test at the 10-percent level of confidence. Using *b* and *ln*(*b*) non of the firms fail this test. For the later period and irrespectively of the definition of the bids, no pair of firms fail this test.

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Table 9. Reduced form bid function

	Cartel period 1995–2001			Post cartel period 2004–2009		
Dependent variable	b_{it}	$ln(b_{it})$	rb_{it}	b_{it}	$ln(b_{it})$	rb_{it}
B_{area}	−36.70*** (3.52)	−0.899*** (0.01)	−0.061 (1.55)	−8.360*** (3.15)	−0.717*** (0.04)	2.601 (5.44)
B_{dens}	0.356** (0.14)	−10.073*** (2.15)	−0.019 (0.06)	−0.498 (2.06)	−17.03 (15.34)	1.194 (3.55)
B_{temp}	0.108 (1.44)	1.272*** (0.37)	0.076 (0.63)	−0.476 (1.96)	0.453 (1.58)	−2.178 (3.37)
B_{rain}	0.042 (0.11)	−0.472*** (0.06)	0.063 (0.05)	−0.475 (0.14)	0.232 (0.24)	0.194 (0.25)
Bids (β_{bids})	11.204 (28.81)	0.002 (0.00)	0.002 (0.00)	1.480 (1.73)	0.004 (0.03)	−0.023 (0.03)
Backlog (β_{blog})	0.789 (0.89)	−26.965 (74.25)	−41.246 (36.95)	9.630 (17.50)	−8.954 (303.20)	1.874 (3.02)
Fixed effects						
Regional	Y	Y	Y	Y	Y	Y
Firm	Y	Y	Y	Y	Y	Y
Year	Y	Y	Y	Y	Y	Y
Observations	1,830	1,830	1,830	299	299	299
R^2_{adj}	0.09	0.77	0.21	0.11	0.68	−0.06
AIC	38,004	3,855	1,296	5,944	764	763

Note(s): Standard deviation within parenthesis. $^*p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$
Source(s): Created by author

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Table 10. The Fisher *z* test statistic for conditional independence during the cartel period (after the cartel period) in the upper part of the matrix

	A	B	C	D	E	F
A						
b	–	15.394 (.)	31.866 (0.296)	14.949 (1.085)	62.674 (5.673)	49.157 (3.737)
ln(*b*)		6.773 (.)	21.786 (4.847)	11.369 (3.111)	36.640 (11.643)	40.468 (12.893)
rwb		−0.512 (.)	3.937 (2.643)	1.770 (−1.715)	9.970 (0.457)	5.675 (0.904)
B						
b	0.574 (.)	–	17.190 (.)	3.471 (.)	15.102 (.)	15.915 (.)
ln(*b*)	0.359 (.)		6.517 (.)	0.525 (.)	7.054 (.)	7.205 (.)
rwb	0.070 (.)		0.788 (.)	−0.050 (.)	0.087 (.)	0.343 (.)
C						
b	0.889 (0.950)	0.703 (.)	–	0.679 (0.237)	30.610 (0.106)	32.981 (1.123)
ln(*b*)	0.808 (0.282)	0.427 (.)		6.683 (1.592)	22.404 (4.657)	21.276 (3.128)
rwb	0.967 (0.999)	0.094 (.)		0.672 (−1.151)	7.658 (0.400)	3.897 (−0.833)
D						
b	0.259 (0.896)	0.166 (.)	0.335 (0.919)		14.776 (0.235)	1.187 (1.466)
ln(*b*)	0.446 (0.868)	0.655 (.)	0.700 (0.590)		11.591 (3.357)	10.734 (3.592)
rwb	0.977 (0.998)	0.392 (.)	0.971 (0.998)		0.701 (1.037)	3.350 (0.741)
E						
b	0.999 (0.983)	0.599 (.)	0.938 (0.998)	0.268 (0.955)	–	48.050 (3.995)
ln(*b*)	0.994 (0.931)	0.376 (.)	0.877 (0.504)	0.408 (0.719)		36.240 (9.844)
rwb	0.477 (1.000)	0.090 (.)	0.950 (1.000)	0.965 (0.998)		6.662 (−0.325)
F						
b	1.000 (0.955)	0.601 (.)	0.864 (1.000)	0.257 (0.955)	0.998 (1.000)	–
ln(*b*)	0.983 (0.988)	0.334 (.)	0.610 (0.369)	0.326 (0.864)	0.814 (0.858)	
rwb	0.977 (1.000)	0.018 (.)	0.967 (0.999)	0.995 (0.997)	0.915 (1.000)	

Note(s): The *F*-test probabilities for exchangeability during the cartel period (after the cartel period) in the lower part of the matrix
Source(s): Created by author

4.4 Lundberg (2017) and Bergman et al. (2020)

Both Lundberg (2017) and Bergman *et al.* (2020) apply spatial econometric techniques to test for conditional independence in bidding behavior across the convicted firms. A positive Moran's I or a positive spatial lag parameter is taken as evidence for collusive bidding behavior. The degree of competition on each contract is controlled for by including the number of bidders in the estimations, which are instrumented by the procurement procedure. Their method allows for testing of conditional dependence across many bidders instead of pairwise testing (Bajari and Ye, 2003) and testing for competitive bidding one firm at a time (Porter and Zona, 1999).

The results presented in Lundberg (2017) and Bergman *et al.* (2020), which are based on the same asphalt data used here, are displayed in Tables 11 and 12 below. The results presented here differs from the results presented in the original papers as we have now introduced the backlog [18]. Results from Moran's I test directly on b_{it} and $ln(b_{it})$ are shown in rows 1 and 2 of Table 11, and the same test applied on the residuals of different specifications of equation (6) are shown in rows 3–8. Irrespective of the specification and if the test is applied on the residuals or directly on the bids, the Moran's I is highly significant for the cartel period. This is in line with collusive bidding. For the postcartel period, the Moran's I is highly significant when applied directly on $ln(b_{it})$ and for the log-linear specification with procurement, firm, region, and year fixed effects are included, while insignificant for all other specifications. This suggests that the higher cartel members' bids are coordinated or at least not independent.

Results from the spatial lag models estimated by Bergman *et al.* (2020) are displayed in Table 12. The spatial approaches can verify a collusive bidding behavior (a positive and significant ρ) during the cartel period and reject such behavior for the later period.

As mentioned above, the use of spatial econometric tools makes it possible to test for collusive bidding behavior across several bidders. While neither the methods suggested by Porter and Zona (1999) nor Bajari and Ye (2003) could verify a collusive bidding behavior by firms *B* and *D*, these two firms are included in the group of collusive firms that did not pass the test of conditional independence based on the spatial approach. If firms *B* and *D* are excluded from the group of cartel firms, the econometric significance of conditional dependence across the remaining firms become stronger. Even though the two firms *B* and *D* where convicted, the results presented here suggest that they were not the main driving forces behind the cartel and they might also have been acting in a more competitive manner compared to the other convicted firms. The main point is that if a single or a few innocent companies are included in the group of suspected firms, the spatial econometric approaches might indicate a collusive bidding behavior across all firms. However, for the asphalt cartel, the spatial technique captures even the bidding behavior by firms *B* and *D* while the other methods do not.

4.5 Our results in relation to other methods and additional interpretations

In a related paper, Aaltio *et al.* (2025) apply the distributional regression approach and the machine learning approach suggested by Clark *et al.* (2025) and Huber and Imhof (2019) respectively on the Swedish asphalt cartel (the same data used in this paper). For the cartel period, they find the losing bids to be clustered and also a gap between the winning bid and the losing bids creating a bimodal distribution of the bids. This is in line with the findings by Chassang *et al.* (2022), Clark *et al.* (2025) and Imhof *et al.* (2018). Aaltio *et al.* (2025) also find the distributional regression approach to correctly reject competitive bidding for the cartel period. However, the machine learning model suggested by Imhof and Huber (2019) provide mixed results. Depending on what training data used, the machine learning model

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Table 11. Moran's *I* statistics (*z* statistics)

Variable/model	Cartel period 1995–2001		Post cartel period 2004–2009		Log-linear	Proc. fixed	Firm. fixed	Region fixed	Year fixed
b_{it}	1.326	(37.444)	0.038	(0.395)	N	N	N	N	N
$ln(b_{it})$	1.104	(30.575)	0.721	(6.317)	Y	N	N	N	N
M1	1.297	(36.596)	0.057	(0.571)	N	N	N	N	N
M2	1.300	(36.662)	0.060	(0.600)	N	Y	N	N	N
M3	1.300	(36.652)	0.048	(0.483)	N	Y	Y	N	N
M4	1.290	(36.342)	0.050	(0.496)	N	Y	Y	Y	N
M5	1.314	(37.007)	0.049	(0.483)	N	Y	Y	Y	Y
M6	0.940	(26.046)	1.078	(9.488)	Y	Y	Y	Y	Y

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Table 12. Parameter estimates

Variables	Cartel period 1995–2001		Post cartel period 2004–2009	
	GMM	GMM (ln)	GMM	GMM (ln)
ρ_{cartel}	0.529*** (8.43)	–	−0.042 (−0.03)	–
ρ_{cartel}(ln)	–	0.019* (2.55)	–	0.034 (1.30)
β_{bids}	48.53 (0.21)	1.766*** (9.55)	58.11 (0.09)	−1.250* (−2.30)
β_{dens}	−258.0** (−2.66)	−6.337** (−2.70)	554.5 (0.14)	−61.59** (−3.38)
β_{area}	−22.16*** (−7.87)	−0.927*** (−61.00)	−8.371** (−2.75)	−0.706*** (−19.21)
β_{temp}	−873.1 (−0.73)	0.493*** (4.27)	−972.1 (−0.24)	−0.480 (−0.50)
β_{rain}	177.6*** (3.79)	1.310*** (3.48)	−12.80 (−0.10)	3.610* (2.47)
β_{blog}	16.61 (0.32)	53.69 (0.80)	−15.10 (0.99)	−16.21 (−0.06)
Fixed effects				
Firm	Y	Y	Y	Y
Regional	Y	Y	Y	Y
Year	Y	Y	Y	Y
Observations	1830	1830	299	299
Pseudo R^2	0.12	0.77	0.23	0.69

Note(s): Standard deviation within parenthesis. *$p < 0.10$, **$p < 0.05$, ***$p < 0.01$
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did correctly classified tenders to be collusive or competitive in 63-percent to 74-percent of the tenders.

To summarize our results and relate them to the findings by Aaltio *et al*. (2025). We find no systematic difference in bidding behavior or the ranking of bids between competitive and collusive bidders in neither period based on the Porter and Zona (1993) approach. We do find a positive correlation between the backlog and participation of cartel firms which suggests that even if loaded with jobs, cartel firms tend to post phony bids to make the impression of competition (Porter and Zona, 1999).

Our results indicate that we can reject the hypothesis of conditional independence (Bajari and Ye, 2003) only among the larger or more frequent bidders. It is reasonable to assume that the most frequent bidders also had the greatest influence over the cartel, which is supported by the court's statement. These companies are also the ones who determine the winner, which means that when the cartel decides that company *A* should submit the lowest bid, company *E* will submit a (somewhat) higher bid. To create the appearance of competition, both *A* and *E* submit bids on virtually all contracts, resulting in many observations and consistently low and higher bids, respectively. Companies *D* and *B* submit few bids and usually high bids because they are seldom designated as the cartel's predetermined winners. Once the cartel has decided which company should submit the lowest bid and the distance between the predetermined winning bid and the other cartel companies' bids, the smaller companies discuss how to place their bids. According to LaCasse (1995), the losing companies do not want to signal that they are not competitive in the private market by submitting excessively high bids. Therefore, the losing companies, primarily the smaller ones *B* and *D*, discuss among themselves how to place their bids. Given this, it also becomes more challenging to demonstrate that *B* and *D* coordinate their bids through the conditional independence test, as they submit relatively few bids. However, the correlation between the nonwinning bids is captured in the spatial model (Lundberg, 2017; Bergman *et al*., 2020) where all nonwinning bids are included as part of the cartel.

Adding backlog, the spatial econometric approaches by Lundberg and Bergman *et al*. still verify collusive bidding behavior during the cartel period and correctly reject such behavior

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after the court order in 2003. This is interesting as it suggests that the approach, at least to some extent, is robust and not sensitive to omitted variable bias. It is also interesting as it provides more information to the findings of a bimodal distribution by Aaltio *et al.* (2025). That is, the phony bids by the nonwinning cartel firms do not only leave a gap to the winning bid, but they are also correlated. Moreover, this result also gives more insights to the pairwise test of conditional independence, especially for the post cartel period. Even though the spatial econometric approach in the case of the Swedish asphalt cartel can verify collusive behavior among the convicted firms, there is a risk in other cases that the spatial analysis may incorrectly indicate that innocent companies are part of a cartel. The main point is that if a single or a few innocent companies are included in the group of suspected firms, the spatial econometric approaches might indicate a collusive bidding behavior across all firms. This could be avoided by applying both methods to the same data or, as done in this paper, excluding the smaller companies from the spatial analysis. However, for the asphalt cartel, the spatial technique captures even the bidding behavior by firms *B* and *D* while the other methods do not. As reported by Aaltio *et al.* (2025), the machine learning approach did correctly classify tenders to be collusive or competitive in between 63-percent and 74-percent of the tenders. This can be compared with Bajari and Ye's test for unconditional dependence. How well their test captures cartel behavior largely depends on the definition used. The prediction rate is between 50-percent and 93-percent for the cartel period and between 10-percent and 90-percent for the postcartel period. Overall, the prediction rate for both the machine learning method and Bajari and Ye's test for conditional independence seems to depend on the training data used and the functional form.

5. Concluding comments

In this paper, the econometric methods proposed by Porter and Zona (1993, 1999), Bajari and Ye (2003), Lundberg (2017) and Bergman *et al.* (2020) have been used to test for collusive bidding behavior on public contracts. Our data set covers the so-called Swedish asphalt cartel where the identity of the colluding firms is known. We also relate our results to the findings by Aaltio *et al.* (2025) who apply a distributional regression approach and the machine learning approach suggested by Clark *et al.* (2025) and Huber and Imhof (2019) respectively on the Swedish asphalt cartel (the same data that is used in this paper).

In our analysis, we find no evidence of a different bidding behavior of the cartel firms as a group compared to the control group based on the method suggested by Porter and Zona (1993) or the ranking of the cartel bids compared to the control group. We did not, based on the methods suggested by Porter and Zona (1999), find evidence that the cartel firms bid more frequently compared to the control group. However, we did find that the smaller cartel firms (in terms of the number of posted bids) tend to bid more in accordance with competitive bidding compared to the larger cartel firms. The test for conditional independence and exchangeability (Bajari and Ye, 2003) suggests that the cartel firms coordinated their bids, a bidding strategy also confirmed by the spatial econometric methods suggested by Lundberg (2017) and Bergman *et al.* (2020).

The different econometric approaches for cartel detection in public procurement data applied in this paper focus on various aspects of how a potential cartel may operate and thereby complement each other and in combination enrichen the analysis and the understanding on the functioning of a potential cartel. This is of importance for public agencies searching for collusive bidding behavior on public contracts on different markets as they are likely to operate differently. As collusive bidding may take many different forms, it is unlikely that all methods applied in this paper will signal collusive bidding behavior on the same cartel as they test for different cartel behavior. While the method Porter and Zona (1993)

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analyze the bids themselves and the ranking of nonwinning cartel bids, the approach by Porter and Zona (1999) focuses on the decision to bid and, given that a bidder participates, what the actual bid is. When Porter and Zona (1999) tested whether the different cartel firms bid differently, it is done for each individual cartel firm. Furthermore, Lundberg (2017) and Bergman *et al.* (2020) build on Bajari and Ye's (2003) premise that noncartel bids should be conditionally independent of each other. Bajari and Ye test for conditional independence between individual bidders, whereas Lundberg and Bergman test for essentially the same behavior but between several cartel firms. Hence, in combination they do complement each other. For instance, even though the ranking of the phantom bids by the undesignated winners among cartel members do not systematically differ from the ranking of competitive bidders (Porter and Zona, 1993), they may still be conditional dependent or correlated (Bajari and Ye, 2003; Lundberg, 2017; Bergman *et al.*, 2020). Hence, cartel firm *i* did not always bid higher than cartel firm *j*, but when cartel firm *i* bid low, cartel firm *j* bid high and vice versa. Given that the cartel is organized in such a way that only the designated winner within the cartel submits a bid, it is not possible to use the methods described by Bajari and Ye, Lundberg, Bergman *et al*., or Porter and Zona (1993). It may also be the case that it is not possible to find statistically significant evidence that nonwinning cartel bids are ranked in a certain way according to Porter and Zona (1999), but that they do correlate with each other (Bajari and Ye, Lundberg and Bergman *et al*.). Bajari and Ye (2003) conduct their test for conditional dependence on a pairwise basis while Lundberg (2017) and Bergman *et al.* (2020) test for correlation between a group of colluding firms.

How can our findings and conclusions be applied by competition authorities in their search to detect unlawful bid-rigging in public procurement? Competition authorities typically work from two different approaches to detect collusive bidding in public procurement. One approach involves following up on tips from the public, competing companies, or their own observations. These tips or observations usually identify specific behaviors among the bidders. Based on such information about how the bidders behave, the competition authority can collect data and choose the method (perhaps one of those presented and evaluated in this paper) that tests for the specifically suspected behavior in that particular market.

The other approach involves the competition authority scanning and testing a variety of markets for potential collusive bidding in their efforts to detect it. Since data and information collection is time- and resource-intensive, it is reasonable for a competition authority to base this type of scanning on the least demanding data collection possible. Therefore, it is also reasonable for these types of investigations to be based solely on the bids. In such case, the proposed methods by Lundberg (2017) and Clark *et al.* (2025), whose distributional approach has also been implemented by Aaltio *et al.* (2025) are useful starting points as they require information on the bids only and no background characteristics.

From the perspective of a competition authority, some considerations should be made before the authority initiates an investigation into a suspected cartel or conducts a more general scan of several different markets. Firstly, a competition authority often has limited resources and must therefore assess in most cases whether the monetary damage caused by the cartel is substantial enough to justify an investigation and the collection of further information. This should involve the direct economic damage that the cartel inflicts on the contracting authority, but also concerns the extent to which the cartel affects the functioning of the market and thereby causes harm to competing companies and consumers. Secondly, are there sufficient procurements to enable an econometric analysis of the companies' bidding behavior? If not, further analysis must be based on other methods and the collection of additional information to substantiate any suspicions. Thirdly, if there is enough data for an econometric analysis, decide on what econometric method is best suited to detect the suspected behavior or scanning.

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Finally, the literature on various methods for detecting collusion in public procurement has recently expanded, and the present study does not claim to be a comprehensive review of all these methods. We leave it to future research to supplement our analysis and apply additional methods to the same procurement data and conditions, such as the Swedish asphalt cartel or other known instances of collusive bidding behavior in public procurement. Or apply the methods evaluated in this paper on other data sets and known cartels.

Notes

1. See, among others, also Feinstein *et al.* (1985), Paarsch (1992), Baldwin *et al.* (1997), Jofre-Bonet and Pesendorfer (2003), Campo *et al.* (2003), Flambard and Perrigne (2001), Abrantes-Metz *et al.* (2006), Kawai and Nakabayashi (2022) and Kawai *et al.* (2023).
2. See for instance Anselin (1988), Anselin and Bera (1998), LeSage and Pace (2009) and Elhorst (2014) for an introduction to spatial econometrics.
3. Compare with medicine where different methods are used to scan sometimes large groups for common and severe deceases which saves both life and resources.
4. Note that the screen suggested by Clark *et al.* (2025) build on the theoretical results of Chassang *et al.* (2022) and Kawai *et al.* (2023).
5. Clark *et al.* (2025) argue that while the cartel might find it optimal to leave a gap between the designated winner and the phantom bids (to make sure the designated winner will get the contract), the cartel is less likely to have the same incentives to manipulate the distance between the phantom bids. The result is a bimodal distribution of bids by members of the cartel.
6. In total, their data set contain nearly 18,000 procurement contracts. Some of the estimations and descriptive statistics presenter in the paper are based on 495 contracts, some on 466, others on 450, 441, 427, 139, 134 and 131 contracts. It's not always obvious on which subsample some of the results are based on.
7. See the seminal work by Anselin (1988), and for more recent texts consider Elhorst (2014), Gibbons *et al.* (2015), Kelejian and Piras (2017), and the references therein.
8. Note that $w_{ii} = 0$.
9. Note that this is related to the test for conditional independence. However, here the test is extended to include more than two bidders and the bid by the designated winner is not included.
10. In 2022, the threshold value for goods and services for municipalities and regions are SEK 2,236,731, for construction contracts SEK 55,991,099 and for social services and another special services SEK 7,802,550.
11. Formally, nine firms were convicted but 3 belonged to the same group of companies and are treated as one firm in our analysis.
12. This data is used and described in detail in Lundberg (2017) and Bergman *et al.* (2020).
13. In order not to single out the convicted companies by name, we choose to use the designations *A*, […], *F*. The interested reader is referred to the court's statement (Stockholms tingsrätt, 2003; Marknadsdomstolens dom, 2003, 2004, 2009) and the summary of the process [Swedish Competition Authority (SCA), 2007, 2009].
14. For a in dept analysis of contract heterogeneity, see Krasnokutskaya (2011). As always, the final results are conditional on these choices.
15. It's far from obvious what the determinants of the phony bids are.
16. Firm fixed effects are not included in the MNL estimations as the model did not converge.

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17. The lowest bid may not meet the standards required by the procurement entity and therefore not reflect the true cost which is the reason why bids are normalized by the winning bid.

18. Note also that the results from the Moran's I test on and $l_n(b_{it})$ presented here also differ slightly to the results presented in Lundberg (2017). This is due to the fact that we have excluded one procurement with only one bidder here.

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